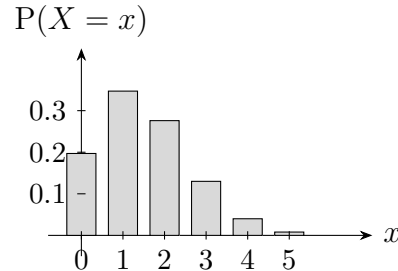
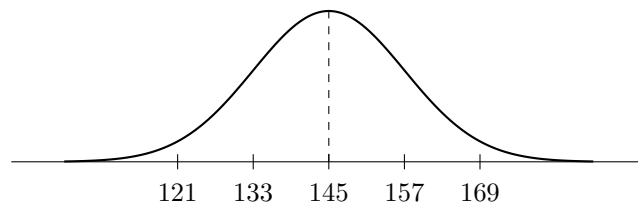


4.18 Normal and Binomial Distribution

1. A bakery finds that 15% of its croissants are overcooked. A box contains 10 randomly selected croissants. Let X be the number of overcooked croissants in the box.



- (a) Find the expected number of overcooked croissants in a box, $E(X)$. [1]
- (b) Find the probability that none of the croissants are overcooked. [2]
- (c) Find the probability that at most 1 croissant is overcooked. [2]
- (d) Find $P(X = 2)$. [2]
- (e) Find $P(X \leq 1)$. [2]
2. The heights of sunflower plants in a garden follow a normal distribution with mean $\mu = 145$ cm and standard deviation $\sigma = 12$ cm.



- (a) Find the probability that a randomly selected plant is taller than 160 cm. [2]
- (b) Find the probability that a plant's height is between 130 cm and 155 cm. [2]
- (c) 80% of the plants are shorter than h cm. Find h . [3]

3. The time T (minutes) for a commuter train journey follows a normal distribution with mean 48 minutes and standard deviation σ minutes. It is known that 5% of journeys take longer than 55 minutes.

(a) Find the value of σ . [3]

(b) Find $P(T < 42)$. [2]

(c) On a particular day, 20 journeys are made. Find the probability that at most 2 of these journeys take less than 42 minutes. [3]

(d) A journey is selected at random. Given that it took less than 52 minutes, find the probability that it took less than 42 minutes. [3]